

# Global cut surgery for tensor-network contraction ordering

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## Abstract

Simulated annealing over contraction trees is the strongest general-purpose optimizer of tensor-network contraction orders, yet late in every run its dominant contraction—the tree’s *waist*—freezes. Across 835 instrumented converged states, this frozen waist is *never* a locally minimal balanced cut: a bounded Fiduccia–Mattheyses pass finds a strictly cheaper cut every time, by 3–12 bits. *Global cut surgery* exploits the blind spot: extract the waist, improve it as a graph cut, cold-rebuild both sides, accept on measured cost. It sets three records on ten benchmarks including the Sycamore circuit, holds the high-quality wall of the time–quality Pareto front, and gains with scale; certified bounds and twenty falsified alternatives show search control alone cannot win.

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## 23 1 Introduction

24 The cost of contracting a tensor network is governed by the order in which its tensors are  
 25 combined. Finding a good order is the computational bottleneck of classical random-circuit  
 26 simulation [1, 2], of quantum error-correction decoding, and of exact probabilistic inference,  
 27 and a rich ecosystem of optimizers has grown around the problem: simulated-annealing tree  
 28 refiners [3], hyper-optimized portfolios over greedy and hypergraph-partitioning construc-  
 29 tions [4, 5], treewidth machinery from the parameterized-algorithms community [6, 7], exact  
 30 methods for small networks [8], and, recently, learned optimizers—reinforcement-learning  
 31 policies [9] and LLM-evolved heuristics [10]. Among these, tuned simulated annealing over  
 32 contraction trees is in practice the strongest general-purpose refiner: as we show below, thir-  
 33 teen mechanically distinct search paradigms and an independent toolchain all converge onto  
 34 its frontier, and on small instances that frontier is *certifiably* locally optimal. Improving on it  
 35 therefore requires attacking a structural weakness, not a better schedule.

36 This paper identifies such a weakness and builds the algorithm that exploits it. The weak-  
 37 ness is the annealer’s *waist*: with the total cost a log-sum-exp over all contractions, the con-  
 38 verged tree’s cost is dominated by its single most expensive contraction, which induces a bi-  
 39 partition of all tensors. Local tree rewrites move one subtree at a time and can cross between  
 40 distinct good bipartitions of the same size only through long, individually uphill move se-  
 41 quences, so late in the schedule the waist freezes. Our motivating measurement (Sec. 4) is  
 42 that this frozen cut is not merely hard to improve locally—it is *never a locally minimal bal-*  
 43 *anced cut of the tensor graph at all*. Across 835 instrumented calls on converged states of  
 44 real benchmark networks, a bounded Fiduccia–Mattheyses pass found a strictly cheaper cut of  
 45 comparable balance every single time.

46 *Global cut surgery* (Sec. 5) turns that observation into an optimizer: extract the waist  
 47 bipartition from the incumbent tree; improve it as a pure graph-cut problem, ignoring the  
 48 tree; if a cheaper comparable-balance cut exists, promote it to the tree top, cold-rebuild each  
 49 side with fixed-work sub-anneals, splice, and accept only if the independently rescored total  
 50 cost drops; iterate on the (possibly moved) waist. An exact rank-non-increasing simplification  
 51 pass—which removes up to 89% of the tensors of real-provenance networks before any search  
 52 begins—runs as preprocessing.

53 We evaluate the algorithm under a hardened protocol (matched single-threaded budgets,  
 54 independent rescoring of every tree from its topology, per-run relabeling, confirmation-based  
 55 records; Sec. 2) and re-measure every headline comparison on a single dedicated machine  
 56 against the reference Julia implementation, its published benchmark suite, and cotengra (Sec. 6).  
 57 Surgery sets records on three of ten instances, its run distribution separates fully from tuned  
 58 annealing’s on surface code  $d=21$ , it holds the entire high-quality wall of the time–quality

Pareto front, and its advantage grows with surface-code distance. Its boundaries are reported with the same care: random 3-regular expanders yield the advantage back at long budgets, and two instances repel it entirely.

Three further results frame the algorithm’s necessity and scope. First, the incumbent must be tuned before being beaten: a single shipped default (an aspirational memory target) costs  $20\times$ – $2100\times$ , more than any mechanism difference we measured (Sec. 3). Second, on small converged instances nothing can win: we certify the annealing frontier width-optimal, localize the optimum to  $[53, 61.5]$  on the circuit proxy, and prove that bounds computable from the isoperimetric profile can go no further (Appendix A)—which is precisely why the gains must come from changing the *input* rather than the search. Twenty falsified alternative mechanisms sharpen the same lesson (Appendix B). Third, on probabilistic-inference workloads the method’s scope splits by structure: annealing-family methods, ours leading, win sparse pedigree networks, while elimination orderings win dense Bayesian networks, and the shipped defaults of TensorInference.jl sit 12–30 bits above the frontier either way (Sec. 7).

## 2 Setup and evaluation protocol

**Instances and cost model.** A tensor network is a hypergraph  $G = (V, E)$ : vertices are tensors and (hyper)edges are indices; contracting to a scalar (or a small open-index set) requires a full binary contraction tree over the tensors. Writing  $\text{cost}_v$  for the  $\log_2$  floating-point cost of internal node  $v$  (the sum of  $\log_2$  dimensions over the union of the operands’ indices), the two standard quality measures are the total time cost

$$\text{tc} = \log_2 \sum_v 2^{\text{cost}_v}, \quad (1)$$

a log-sum-exp over all  $|V|-1$  contractions, and the space cost  $\text{sc}$ , the  $\log_2$  size of the largest intermediate tensor. Minimizing (1) is NP-hard [11], a hardness discovered independently three times: for join ordering in relational databases [12], for sum-of-products loop optimization in compilers [13], and for tensor networks—and algorithms migrate across these communities, most recently join-ordering algorithms into tensor networks [14]. All numbers in this paper are  $\log_2$  quantities; a difference of 1.0 is a factor of two in flops. Equation (1) is the root of everything that follows: because  $\text{tc}$  is a log-sum-exp, the converged tree’s cost is carried almost entirely by its most expensive contraction—the *waist*.

Table 1 lists the ten benchmark instances. Two are closed synthetic networks used for the certification study (a 250-tensor random 3-regular graph and a 561-tensor Sycamore-like circuit proxy, which we distinguish explicitly from the real circuit). The remainder are imported from the OMEinsumContractionOrders benchmark suite [15] with provenance recorded at import: the real Sycamore 53-qubit, 20-cycle network, surface-code syndrome networks at distance  $d = 9 \dots 21$ , an independent-set counting network on a king graph, a 1000-vertex random 3-regular graph, a 97-qubit random-circuit network, the  $n = 28$  queens counting problem, a deep-belief-network inference instance, and a 27-qubit quantum Fourier transform. Section 7 adds ten hard instances from the UAI-2014 inference competition, exported directly from the artifact shipped with TensorInference.jl.

**Protocol.** Every optimizer run gets 90 seconds of single-threaded wall clock per instance unless stated otherwise. Emitted trees are rescored by an independent checker that recomputes  $\text{tc}$  and  $\text{sc}$  from the tree topology and the original index sets alone; optimizer-reported numbers are never trusted. Instances are randomly relabeled per run, so memorized answers keyed on input encoding do not transfer. A best-known value (“record”) only moves when a run

| instance           | tensors  | indices   | provenance                                    |
|--------------------|----------|-----------|-----------------------------------------------|
| reg3_250           | 250      | 372       | synthetic, closed (certification study)       |
| sycamore_m20       | 561      | 963       | synthetic Sycamore-like proxy (certification) |
| sycamore_53_20_0   | 3369     | —         | real Sycamore 53q, $m=20$ [4]                 |
| surfacecode_d9-d21 | 219–2203 | —         | surface-code syndrome networks                |
| ksg                | 5197     | —         | king-graph independent-set counting           |
| reg3_1000          | 1000     | 1500      | random 3-regular, $n=1000$                    |
| rqc_97_m24         | 1238     | —         | 97-qubit random circuit                       |
| nqueens_28         | 4252     | —         | 28-queens counting                            |
| dbn_13             | 572      | 44 labels | deep-belief-network inference (UAI DBN_13)    |
| qft_27             | 405      | —         | 27-qubit QFT, 27 open indices                 |

Table 1: Benchmark instances. The first two are the closed synthetic networks of the certification study (Appendix A); the rest are imported real-provenance benchmark networks. Ten additional UAI-2014 inference instances are introduced in Sec. 7.

beats the incumbent by more than 0.05 and a second run under fresh relabeling confirms; the worse of the two runs is recorded. On instances above 1000 tensors, where single-run variance exceeds method differences, scored values are medians of three runs. Per-child CPU-time accounting rejects hidden parallelism. Reference rows (the tuned annealer of Sec. 3, and cotengra’s hyper-optimizer and SA refiner [4]) are measured best-of-three with the identical pipeline.

**Same-machine re-measurement.** Because the record board accumulated on a development machine, all headline comparisons were re-measured from scratch on a single dedicated 2-core Linux server: 305 scored runs covering budget scaling (90/300/900 s, three repetitions), run distributions (15 repetitions), and the surface-code family (five repetitions), plus a matched-budget ladder of the reference Julia implementation (OMEinsumContractionOrders.jl: TreeSA at both memory-target settings and one/four/eight restarts, HyperND, and three treewidth back-ends) run through that benchmark suite’s own harness, and the inference batch of Sec. 7. Every cross-method statement in Secs. 6 and 7 is same-machine and matched-budget.

### 3 Tuning the incumbent: a prerequisite

Beating an untuned incumbent is not evidence of algorithmic progress, and the incumbent here ships untuned. The TreeSA-family annealers [3] optimize a composite objective in which the space cost is pulled toward a target  $sc_{\text{target}}$ , with a penalty active whenever  $sc > sc_{\text{target}}$ . The implementation we study (omeco, a Rust re-implementation aligned with OMEinsumContractionOrders.jl) inherited the shipped default  $sc_{\text{target}} = 20$ . On instances whose good trees have naturally larger intermediates—the optimum-quality trees on our two closed instances have  $sc = 34$  and 53—this default is not merely unhelpful. It actively drags the search into a low-memory, high-flop region of tree space.

Figure 10 (Appendix C) shows the resulting  $tc$  at a fixed 90-second budget as a function of  $sc_{\text{target}}$ . The transition it reveals is a cliff, not a slope: every target below the natural space cost of good trees is roughly equally harmful (medians 45.6–47.0 on reg3\_250, 71–77 on sycamore\_m20), quality snaps to the frontier exactly at the natural value ( $sc_{\text{target}} = 34$  and 53), and stays there unchanged through  $sc_{\text{target}} = \infty$ , i.e., with the memory objective deleted. The harm is strictly one-directional. At the shipped default the damage is  $4.3 \log_2$  units on reg3\_250 and 9.7 on sycamore\_m20 ( $20\times$  and  $800\times$ ); on the real Sycamore net-

work it reaches 11.1 units (2100×, Sec. 6). The cliff is visible in *published* data as well: the benchmark suite of OMEinsumContractionOrders.jl reports Julia TreeSA at  $tc = 71.0$  on the real Sycamore network with its shipped  $sc_{\text{target}} = 20$ , within 1.6 of our re-measurement of the same configuration, and 11 units above the same annealer with the target removed.

The practical rule is: *never anneal toward a memory target below the natural space cost of good trees; absent a physical memory budget, use the natural value or no target at all.* Every comparison in the remainder of this paper is against the annealer with this fix applied—the *tuned reference*.

**Why not better search control.** Before targeting the waist, we verified that no search-control change can win where the tuned annealer converges: thirteen mechanically distinct search paradigms and an independent toolchain (cotengra) all land within 0.35 of its frontier on the two closed instances, the frontier is certifiably width-optimal with the optimum localized to  $[53, 61.544]$  on the circuit proxy, and we prove that every lower bound computable from the isoperimetric profile is capped below the balanced-cut value. The full certification — the thirteen-mechanism convergence, two sum-form bound theorems, and the impossibility result — is given in Appendix A, and twenty further falsified mechanisms are taxonomized in Appendix B. The lesson they encode is uniform: global information helps only when injected as *input*, never as search control. The remaining question is where such an input exists, and the answer is the waist.

## 4 The frozen waist is never a minimal cut

Every converged annealed tree has a waist: by (1) its cost is dominated by the argmax-cost contraction, whose leaf set  $A$  (against complement  $B$ ) is a bipartition of all tensors, and whose cost is essentially the weight of the cut  $(A, B)$  in the tensor hypergraph. Local tree moves—the associativity rewrites of the annealer [3]—transfer one subtree across this bipartition at a time; crossing to a *different* good bipartition of comparable balance requires a long sequence of individually uphill steps, which a cooled annealer will never take. We also know the freeze is not repairable by small exact windows: exhaustive splicing of all subtrees up to 16 leaves leaves frontier trees unchanged (they are window-exact optimal, Appendix B), while the dominant contraction spans a constant fraction of the network.

The question is whether the frozen waist is actually good *as a cut*. We instrumented every surgery call of the algorithm of Sec. 5 on its two record runs: at each converged incumbent, extract the waist bipartition and run a bounded Fiduccia–Mattheyses (FM) [16] search for a cheaper cut of comparable balance on the tensor hypergraph, recording the incumbent’s cut weight, the best alternative found, and the structure of the improved cut.

Figure 1a shows all 835 pooled calls on two instances, and **this panel is the paper’s motivating result**: every single point lies strictly below the diagonal. *The annealer’s waist is never a locally minimal balanced cut*—bounded FM finds a strictly cheaper cut of comparable balance on every call, by margins of 3–12 bits. The improved cuts are sparse rather than near-clique (clique fraction 0.01–0.05): genuine alternative separators, not degenerate rearrangements of the same one. The blind spot is structural and persistent: it does not shrink as the incumbent improves, and it recurs at the new waist after every accepted repair. An optimizer that can see cuts globally therefore always has a move available exactly where the annealer is most stuck.

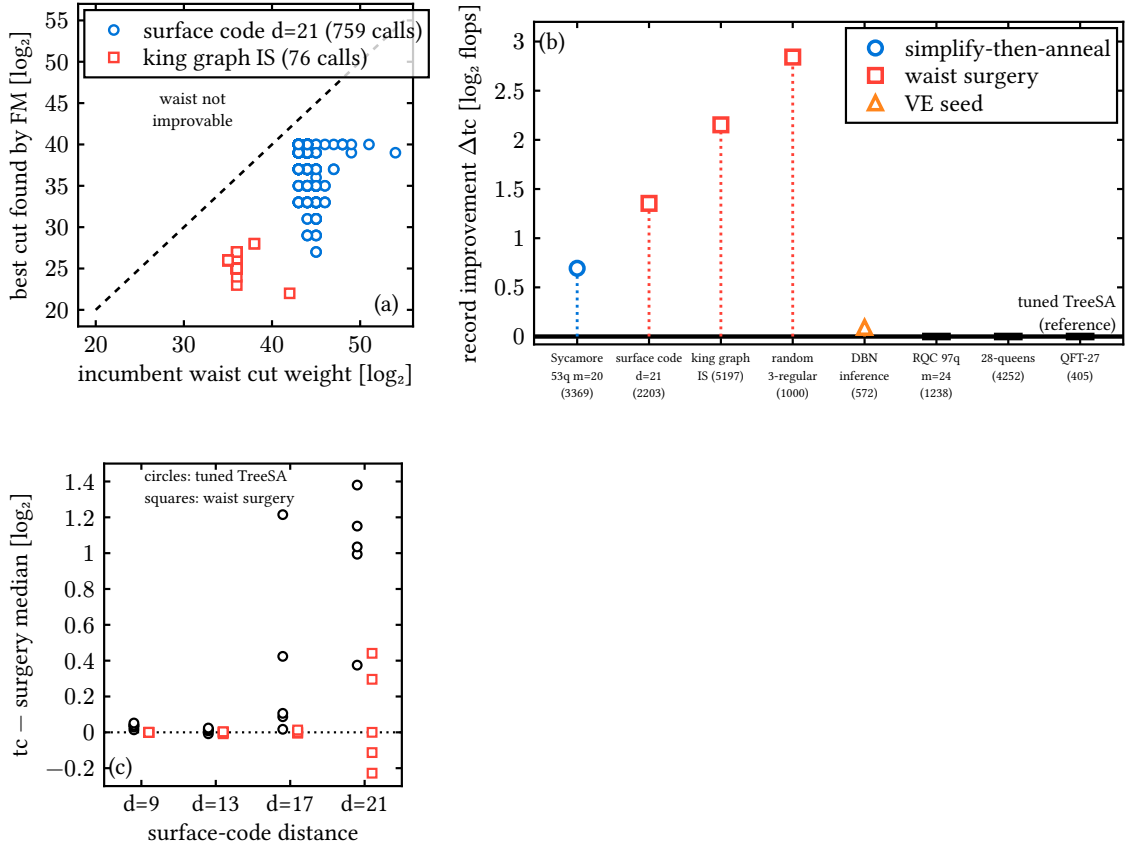


Figure 1: The blind spot and the advantage it buys. (a) Every surgery call on the two record runs: incumbent waist cut weight versus the best comparable-balance cut found by bounded FM. All 835 points lie strictly below the diagonal—the annealed waist is never locally minimal (Sec. 4). (b) Record board: confirmed improvement over the tuned TreeSA reference on each instance (validator protocol: confirm-twice, worse-of-two, per-run relabeling); marker shape encodes the mechanism, thick dashes mark instances where the tuned reference remains the record holder. (c) Surface-code family: all repetitions at 90 s versus code distance, anchored at the surgery median per distance—the surgery advantage grows with  $d$  while its run-to-run spread stays collapsed.

## 174 5 The algorithm: global cut surgery

175 Global cut surgery operationalizes the observation of Sec. 4; Fig. 2 illustrates both of the  
 176 pipeline’s mechanisms on a toy network. The optimizer maintains an incumbent contraction  
 177 tree produced by a standard annealing engine, and interleaves, at every outer iteration, a  
 178 surgery step that treats the waist as a graph problem:

- 179 1. **Extract.** Locate the argmax-cost node of the incumbent (costs recomputed exactly from  
 180 the index dimensions); its leaf set  $A$  versus complement  $B$  is the waist bipartition.
- 181 2. **Improve the cut, not the tree.** Run bounded Fiduccia–Mattheyses passes on the ten-  
 182 sor hypergraph, with gain defined as the exact reduction in summed  $\log_2$  dimensions  
 183 of the straddling labels (open output labels are excluded—they are invariant), and bal-  
 184 ance constrained to  $|A| \pm 18\%$ . The search is seeded from the incumbent cut plus four  
 185 boundary-BFS alternative bipartitions (one from the highest-degree tensor, three from

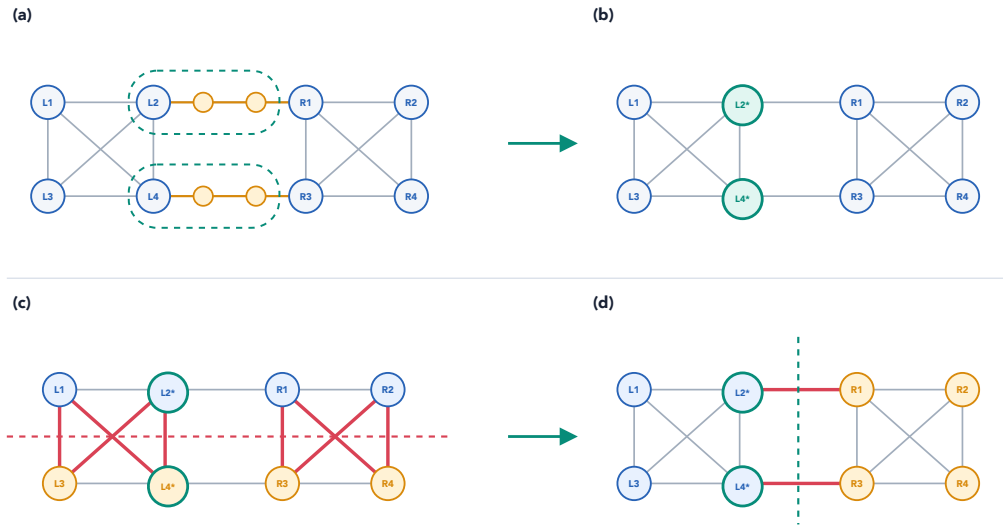


Figure 2: The two mechanisms on a toy network: two dense clusters (each a  $K_4$ ) joined by two sparse bridges. (a,b) Preprocessing: the chain tensors on the bridges are rank-non-increasing, so the exact simplification pass absorbs them ( $L_2, L_4 \rightarrow L_2^*, L_4^*$ ) before any search begins. (c,d) Surgery: a converged annealed tree can freeze with a poor waist—the horizontal bipartition (c), which cuts through both clusters—while the cheap comparable-balance cut through the two bridges (d) is invisible to local tree moves; bounded FM on the tensor graph finds it, and both sides are rebuilt cold and spliced.

random starts), so it can jump to a different separator rather than only polish the current one.

3. **Rebuild cold, with fixed work.** If a strictly cheaper cut of comparable balance was found, promote it to the tree top: each side becomes an independent, much smaller ordering problem (its open indices are the new cut plus its external labels, computed by outside-occurrence counting so the rebuilt tree’s top node matches the independent scorer exactly). Anneal each side from scratch with a *fixed-work* budget (three V-cycles, linear schedules), join, and splice into the incumbent.

4. **Accept on measurement.** Adopt the rebuilt tree only if its independently recomputed total cost improves; then iterate—the waist typically moves, and surgery repeats at the new waist until no improving cut is found or the budget ends.

Two design choices matter beyond the loop itself. *Fixed work, not fixed time:* giving each rebuild a deterministic amount of annealing work (rather than a wall-clock slice) collapses run-to-run spread from 0.8 to 0.08 bits—the algorithm’s variance behavior is set by the rebuild policy, not by the machine. *Surgery needs its base engine, and vice versa:* an ablation that keeps only the ratchet base engine plateaus far above the full algorithm (surface-code  $d=21$ : 49.6 ratchet-only versus 47.38 with surgery; king graph: 50.0 versus 36.8), while surgery without a competent annealed incumbent has nothing worth repairing. All knobs scale with instance size; nothing is tuned per instance.

*Relation to prior refiners.* Refinement layered on top of a construction is an active line: the TreeSA family itself [3], simulated annealing over partition assignments [17], and, closest to this work, nearest-neighbor-interchange refinement of hyper-optimizer output, whose gain

over cotengra grows with bond dimension [18]. All of these move the *tree* locally. Surgery differs in kind: it improves the dominant *cut* as a graph problem and hands the result back as input to a rebuild—and the measurement of Sec. 4 is precisely that the local refiners’ fixed point leaves this cut improvable every time.

## 5.1 Preprocessing: exact simplification

Real-provenance networks carry large amounts of *rank-non-increasing* structure: tensors whose contraction with a neighbor can never increase any intermediate rank (rank-1 spiders, dangling legs, series chains). Preprocessing of this kind is established practice (cuTensorNet’s pathfinder simplifies before partitioning [19], and opt\_einsum removes single-tensor reductions [20]); our pass is the exact rank-non-increasing closure with splice-back, so it composes with the record protocol. Before any search, the pass contracts this structure away greedily; the optimizer runs on the reduced network, and the contracted fragments are spliced back into the final tree as fixed subtrees. The pass is exact and costs milliseconds.

Its effect is largest exactly where synthetic benchmarks never look (Fig. 11a, Appendix C): 88.7% of the real Sycamore network’s 3369 tensors are removed before annealing begins ( $3369 \rightarrow 381$ ), 65.5% of surface-code  $d=21$ , 33.9% of nqueens<sub>28</sub>—and 0% of the random 3-regular control, which is why the identical mechanism, correctly measured as useless on the pre-simplified synthetic proxy two cycles earlier, was nearly discarded. A matched-budget ablation on the real Sycamore network isolates the contribution: the same anneal reaches  $tc = 61.3$  with simplification and 76.0 without it, a  $2^{14.7}$  ( $\sim 27000\times$ ) difference from preprocessing alone. Figure 11b shows the resulting record march: three successive simplify-then-anneal implementations drove the Sycamore record  $60.605 \rightarrow 60.169 \rightarrow 60.104 \rightarrow 59.911$ .

## 5.2 Seeding and anytime behavior

The pipeline emits its incumbent eagerly (atomic writes), so it is an anytime optimizer, and its construction stage is pluggable. Two components earn their place. *Elimination seeds*: on hypergraph-like instances with few, high-degree labels, seeding the annealer with a min-fill variable-elimination order gives it a start ( $tc = 28.8$  at 0.2 s on the DBN instance) that greedy construction (44.7) cannot approach; the mechanism’s boundary is equally sharp—at 4086 labels the VE peel freezes and the seed is useless. *Racing*: arms with budget-fraction linear schedules, composed by a router, produced three records in a single run during development; a probe-leader policy never won. Harness-clocked time-to-frontier records exist as well (16.1 s to the reg3<sub>250</sub> frontier versus 20.2 s for the reference; 0.2 s on the DBN instance).

## 6 Evaluation

**Records.** Figure 1b assembles the campaign’s confirmed records against the tuned reference on all ten instances. Global cut surgery holds three (king graph 36.96, surface code  $d=21$  47.40, reg3<sub>1000</sub> 131.07—the only record on an expander in the entire campaign), the simplify-then-anneal line holds the real-Sycamore record (59.91), and the elimination seed holds the DBN record (29.00). On four instances—both certified closed instances, the 97-qubit random circuit, and 28-queens—the tuned reference itself remains unbeaten, exactly as the certified null of Appendix A predicts for the converged cases. A representative surgery run (Fig. 9, Appendix C) shows the dynamics: each accepted rebuild drops the incumbent below the pre-surgery record, and the official median lands at 47.377.

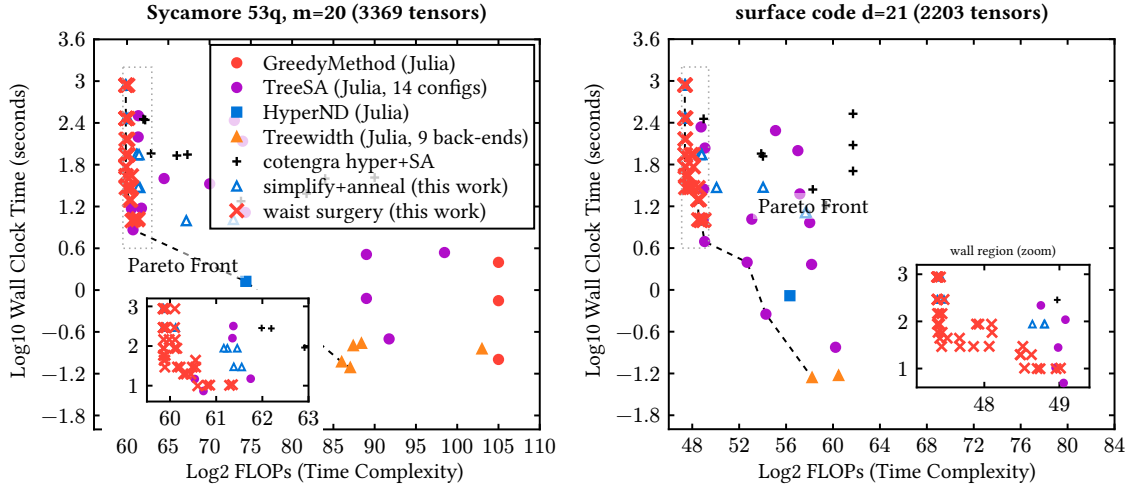


Figure 3: Time-versus-quality Pareto view of the same-machine campaign (dedicated 2-core server, all runs single-threaded), in the presentation style of the OMEinsumContractionOrders benchmark study [15]. Each point is one optimizer configuration or repetition. Baselines are the published implementations: Julia TreeSA across a 14-configuration ladder (both memory-target settings, 1–8 restarts, five sweep depths), GreedyMethod, HyperND, nine treewidth back-ends, and the cotengra hyper-optimizer with SA refinement at four budgets. Open blue triangles and red crosses are this work, at budgets from 10 to 900 s. Our tuned TreeSA port (not shown) coincides with the Julia TreeSA points wherever configurations overlap. The dashed line is the non-dominated front. Global cut surgery (crosses) holds the entire high-quality wall of the front on both instances; the insets zoom the wall region (dotted window), where the separation is explicit: on Sycamore the surgery front sits at  $tc = 59.85\text{--}59.9$  from 60 s on (first reaching 59.89 at 30 s) while every TreeSA point stays at 60.5–61.8 at *all* budgets, and on surface code surgery reaches the 47.4 floor at 30 s while no TreeSA configuration gets below 48.75 at any measured budget. Orderings beyond the axis range (BFS/MCS-style treewidth, up to  $tc = 443$ ) are omitted.

**The time–quality Pareto front.** Figure 3 presents the same-machine campaign as a Pareto scatter, following the presentation of the OMEinsumContractionOrders benchmark study [15]: every optimizer configuration from every toolchain is one point, and the dashed line traces the non-dominated front. On both instances the front has the same anatomy: treewidth back-ends hold the fast/low-quality end (sub-second, 10–40 bits off best), HyperND the middle, and *global cut surgery* holds the entire high-quality wall—at every time tier (90/300/900 s) its points sit strictly left of every tuned-TreeSA point from either implementation. On the real Sycamore network surgery reaches 59.87 at 90 s while the tuned reference does not reach it even at 900 s (60.50)—a quality gap, not a speedup; on surface-code  $d=21$ , surgery at 90 s (47.39) matches the reference at 900 s (47.38)—a tenfold time-to-quality gain with both converging to the same floor.

**Distributions and scaling.** Figure 12 (Appendix C) gives the full run distributions. Over 15 repetitions at 90 s, the surgery and reference distributions on surface-code  $d=21$  are fully separated—surgery’s worst run (47.91) beats the reference’s best (48.18); king-graph and reg3\_1000 medians improve by 3.0 and 2.4; the elimination-seed mechanism shows a 14× tighter spread than the reference on the DBN instance. On the surface-code family the surgery

| instance         | this work     | Julia OMECO suite |                   | cotengra |
|------------------|---------------|-------------------|-------------------|----------|
|                  | record (90 s) | best $\leq 90$ s  | best $\leq 900$ s | SA, 90 s |
| sycamore_53_20_0 | <b>59.91</b>  | 64.49             | 61.35             | 66.03    |
| surfacecode_d21  | <b>47.40</b>  | 48.98             | 48.75             | 54.03    |
| ksg              | <b>36.96</b>  | 39.22             | 38.27             | 69.04    |
| reg3_1000        | <b>131.07</b> | —                 | —                 | 165.21   |
| dbn_13           | 29.00         | <b>28.03</b>      | <b>28.03</b>      | 41.15    |
| qft_27           | <b>29.61</b>  | 29.62             | 29.62             | 31.45    |

Table 2: Records versus external baselines, all measured on the same server. Julia columns: over the configuration ladder of the reference implementation (OMEinsum-ContractionOrders.jl: TreeSA at  $sc_{\text{target}} \in \{20, \infty\}$  with 1/4/8 restarts, HyperND, treewidth back-ends, run through its own harness), the best row whose measured runtime fits budget  $B$ . Cotengra column: median of three 90 s single-threaded runs of the hyper-optimizer followed by subtree simulated annealing; under this protocol it is starved of the parallel trial sampling that is its main strength, so these rows are a protocol-matched reference, not a claim about cotengra at large. Bold marks the winner per instance. The dbn\_13 row is won by the treewidth back-end (min-fill), instantly—see Sec. 7. reg3\_1000 is not in the Julia suite. nqueens\_28 is excluded: no ordering is stable at these budgets (see text).

266 advantage grows with distance (+0.04, +0.02, +0.11, +1.04 at  $d = 9, 13, 17, 21$ ; Fig. 1c)  
 267 while its run-to-run spread stays collapsed: global cut repair pays off precisely at scale, where  
 268 the annealer’s frozen waist is largest.

269 **External baselines.** Table 2 closes the loop against the reference Julia implementation, run  
 270 on the same server through its own benchmark harness at a ladder of configurations, against  
 271 the suite’s published numbers, and against cotengra. Our records beat the best same-machine  
 272 Julia configuration *at ten times our budget* on the three large structured instances (Sycamore  
 273 59.87 versus 61.35; surface code 47.40 versus 48.75; king graph 36.96 versus 38.27), and beat  
 274 the entire published multi-optimizer suite by 6.8, 4.9, and 2.0 bits respectively. The cotengra  
 275 column adds a protocol-matched cross-toolchain reference: on the small converged instances  
 276 it confirms the shared frontier (reg3\_250 40.5 versus our 39.9; the circuit proxy 62.4 versus  
 277 61.5), while on the large instances a single-threaded 90-second budget starves its sampling-  
 278 based portfolio (69.0 on the king graph, 187 on the 97-qubit circuit), which quantifies how  
 279 much of that toolchain’s power lives in parallel trial throughput rather than in per-trial quality.

280 **Where the advantage ends.** Honest boundaries, all measured. On reg3\_1000 the reference  
 281 overtakes surgery at 900 s (129.8 versus 134.4): expanders have no cheap alternative separa-  
 282 tors for FM to find, so the surgery advantage there is a fixed-budget effect. On the DBN  
 283 instance the published (and reproduced) treewidth row wins outright (28.03, instantly, versus  
 284 our 29.00)—elimination methods are the right tool for dense hyperedge-structured networks,  
 285 a theme Sec. 7 develops. On nqueens\_28 no stable ordering exists at these budgets: our  
 286 reference’s fifteen-run spread on the server spans 104.9–159.3, single Julia runs land inside  
 287 that spread, and we decline to claim a direction. The Sycamore simplify-then-anneal pipeline,  
 288 tuned against a faster development machine, is 0.5 *worse* than the reference at 90 s on the  
 289 slower server and recovers only at 300 s—optimizer pipelines calibrate to absolute time, and  
 290 cross-machine reporting must say so.

## 7 Application: inference workloads

Contraction ordering is not a quantum-only problem: exact probabilistic inference contracts the tensor network of a graphical model, and TensorInference.jl [21] does so literally, optimizing an einsum whose inputs are one unity tensor per variable plus one tensor per factor, at full cardinalities. The package’s own cross-tool evaluation [22] shows it competitive with dedicated inference engines (Merlin, libDAI, JunctionTrees) on the same networks, so its end-to-end runtime is bounded directly by the ordering quality studied here. We exported all 65 instances of the UAI-2014 MAR benchmark from the package’s own artifact in exactly that form, ranked them by a 5-second hardness scan, and measured the ten hardest (three dense deep-belief networks, four pedigree-linkage networks, one CSP, one grid, plus the package’s own benchmark instance Promedus\_14) under the full protocol: our methods at five repetitions each, against a same-machine Julia ladder that includes the package’s two defaults (the signature default GreedyMethod and the benchmark configuration TreeSA(ntrials=1, niters=5)), HyperND, the min-fill treewidth back-end, and tuned TreeSA.

Figure 4 shows every method’s distance above the per-instance frontier. Three findings. *First, the shipped defaults are 12–30 bits off frontier* ( $4000\times$  to  $10^9\times$  in flops): GreedyMethod lands 14.6–16.6 above on the DBN family and 30.5 above on linkage\_15; the benchmark TreeSA configuration is little better there (10–16 above). Whatever mechanism one prefers, an inference user who accepts defaults pays orders of magnitude for it. *Second, the frontier splits into two structural regimes.* On dense DBNs, elimination methods win outright: min-fill treewidth and HyperND reach the frontier (25.9–28.0) in milliseconds, tuned annealing trails by 5–8 bits, and our elimination-seeded annealer (Sec. 5.2) lands within 0.7 of the elimination frontier with a  $10\times$  tighter spread than the plain reference. On pedigree linkage the regimes invert: min-fill is 6–15 bits above the frontier, HyperND’s back-end (KaHyPar) segfaults on all four instances, and annealing-family methods—ours leading on all four—hold the frontier. *Third, the regime boundary is a structure property, not a size property:* the DBN networks have 44–66 shared labels of high degree (elimination’s sweet spot), while linkage networks are sparse with a thousand variables. The hard families are not a quirk of our selection: an early triage in the package’s own issue tracker found 105 of the 142 UAI-2014 instances failing, and explicitly asked whether the contraction-order optimizer was leaving the DBN and pedigree families far from the treewidth optimum [23]. The reference tree decompositions posted there (Tamaki’s solver) let us answer that question with an external anchor: on all four dense instances in our hard set the best elimination-method space cost equals the reference treewidth exactly (sc = 21/20/22/20 against largest bags 22/21/23/21 on DBN\_12/14/16 and Grids\_15), so the 12–30-bit default gap is measured against a certified-width frontier, not merely against the best order we happened to find. A portfolio that probes elimination first (milliseconds) and falls back to the annealing pipeline otherwise captures the frontier everywhere we measured; the simplification pass (Sec. 5.1) additionally absorbs the per-variable unity tensors that inference-style einsums carry by construction.

## 8 Conclusions

We introduced global cut surgery, a contraction-order optimizer built on a structural measurement: the converged annealer’s dominant contraction—the cut that carries its cost—is never a locally minimal balanced cut of the tensor graph (835 instrumented calls, zero exceptions, margins of 3–12 bits). Surgery extracts that cut, improves it globally with bounded Fiduccia–Mattheyses passes, cold-rebuilds both sides with fixed-work sub-anneals, and accepts on measured cost. Together with an exact simplification preprocessing pass, it sets records on real

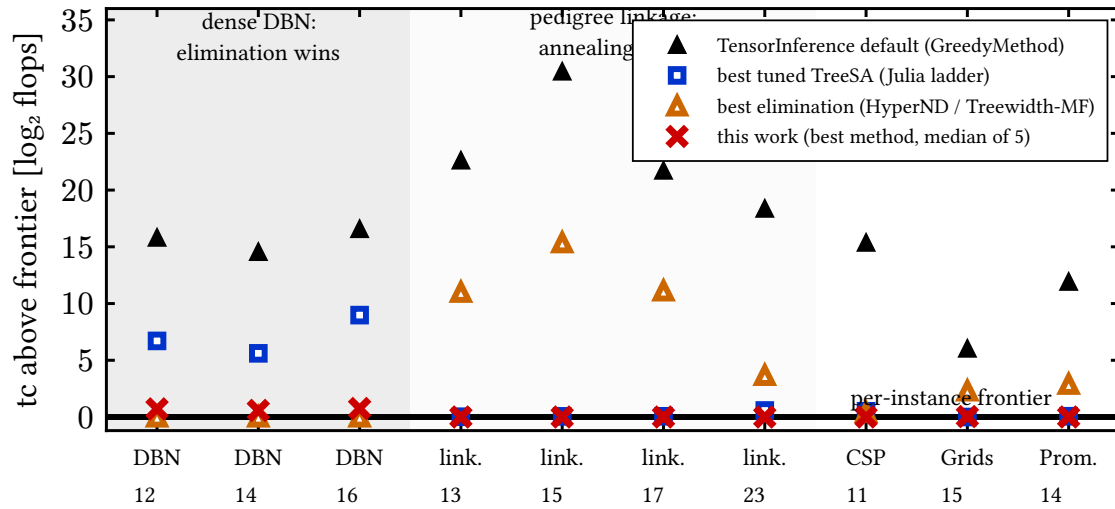


Figure 4: The ten hardest UAI-2014 MAR instances (from a 65-instance scan): each method’s tc above the per-instance frontier, same machine, 90 s budgets for search methods. Filled triangles: TensorInference.jl’s signature default. Squares: best tuned TreeSA from the Julia ladder. Open triangles: best elimination method (min-fill treewidth / HyperND; KaHyPar segfaults on all four linkage instances, so its rows there are min-fill only). Crosses: this work (best method, median of five). Shaded bands mark the two regimes.

quantum-circuit, surface-code, and counting networks; separates fully from tuned annealing’s run distribution; holds the entire high-quality wall of the same-machine time-quality Pareto front against the reference Julia implementation, its published suite, and cotengra; and its advantage grows with problem scale.

The algorithm’s design is uniquely constrained by negative results: on converged instances the annealing frontier is certifiably width-optimal (optimum in  $[53, 61.5]$  on the circuit proxy, with profile-based bounds provably exhausted), and twenty falsified mechanisms show that global information helps only as input, never as search control. The method’s boundaries are mapped: expanders return the advantage at long budgets, dense inference networks belong to elimination orderings (our elimination-seeded variant comes within 0.7 bits), sparse pedigree networks belong to annealing—and shipped inference defaults sit 12–30 bits above either regime’s frontier, so a milliseconds-cheap elimination probe belongs in front of every annealer.

Deliverables: the surgery pass, the simplification pass, tuned defaults, a warm-start annealing API, and two bug fixes are released in the omeco library (prepared for upstreaming to OMEinsumContractionOrders.jl), together with the validator, all certificates, and the 56-attempt audit trail. Open problems: carving-width-type lower bounds; the two instances that repelled every mechanism (the 97-qubit random circuit and 28-queens, the latter unstable at practical budgets); heterogeneous bond dimensions; and sliced contraction, where the interaction between slicing and surgery is unexplored.

## Acknowledgements

The author thanks the OMEinsumContractionOrders.jl and TensorInference.jl communities for the open benchmark suites and artifacts this study builds on.

**Code and data availability** The optimizer library (omeco), the validation harness, all certificates (treewidth minors, cut sets, and their verification scripts), per-attempt logs, cycle reports, the same-machine campaign data (305 + 200 + 80 + 69 + 30 scored runs and the Julia ladder), and the UAI instance exporter are available in the project repository. Every figure regenerates from JSON data files via Typst sources shipped with the manuscript.

**Use of AI systems** This research was conducted as a steered autonomous research campaign: LLM-based agents proposed hypotheses, implemented attempts in isolated worktrees, and drafted analyses, under human steering decisions documented verbatim in Appendix E and under a validation protocol in which no agent-reported number is trusted—every scored quantity is recomputed by an independent checker from artifacts, and all certified claims are verified by machine-checkable certificates independent of any agent output. The manuscript was drafted with the same agents and revised by the human author, who takes full responsibility for its content.

**Funding information** To be completed before submission.

## A Why search control cannot win: a certified null

The case for operating on the input rather than the search is made by a null result, and we certify it. On the two closed instances the tuned annealer converges within the budget, and nothing we or an independent toolchain tried can beat it—for reasons that are provable, not anecdotal.

### A.1 Thirteen mechanisms, two toolchains, one frontier

We attempted to beat the tuned reference with mechanically distinct search paradigms, implemented independently and scored under the protocol of Sec. 2: coarse subtree-transfer moves and basin hopping [17], evolutionary replica populations, exact-penalty ramps, tabu search with strategic oscillation, large-neighborhood destroy-and-repair, beam search with lookahead, Boltzmann-greedy portfolios [4, 24], recursive balanced min-cut construction [5, 25], elimination-order annealing [6], order-then-tree interval dynamic programming [26], hierarchical coarsen-then-exact solves, structure-guided sweeps, and profile-aware annealing.

Figure 5 shows every scored result as its distance from the best verified value. All thirteen mechanisms land within  $0.35 \log_2$  of the tuned reference on both instances—most within noise—and none set a confirmed record. An independent toolchain reproduces the same frontier: cotengra’s simulated-annealing refiner [4] ties it to within 0.05 and 0.14, while cotengra’s hyper-optimization mode alone remains 1.7 and 2.4 behind at matched budget. Extending the budget tenfold moves the frontier by less than 0.1. At this scale the frontier is a property of the cost landscape, not of any search mechanism, schedule, or implementation.

### A.2 Certified bounds: width-optimality and an impossibility result

Every classical lower-bound technique for contraction cost is *max-form*: it bounds the single largest contraction. The Markov–Shi correspondence [1] equates the minimal largest contraction rank with the treewidth of the line graph  $L(G)$ , and any balanced edge of a contraction tree yields a cut whose boundary lower-bounds the largest contraction.

On sycamore\_m20 these tools give a sharp result. An explicit temporal cut severing all 53 qubit world-lines gives a balanced cut of value 53; extensive search finds nothing smaller,

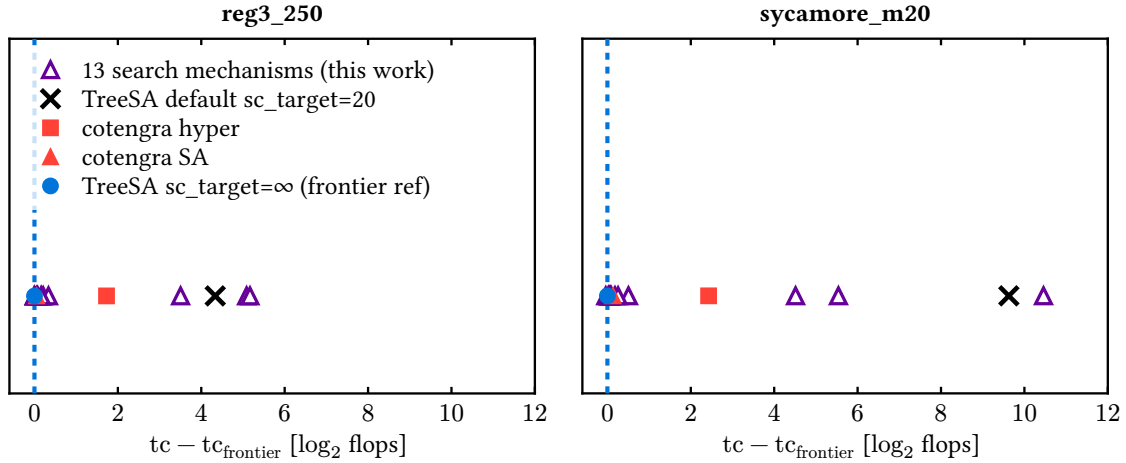


Figure 5: Distance of every scored optimizer from the frontier value (dashed line at zero). Open triangles: the thirteen search mechanisms of this work. Filled symbols: the tuned TreeSA reference ( $sc_{\text{target}} = \infty$ , circle), cotengra’s SA refiner (triangle), cotengra’s hyper-optimizer (square), and the shipped-default TreeSA (cross).

400 and the frontier trees achieve  $sc = 53$  exactly. The frontier is therefore *width-optimal*, and  
 401  $tw(L(G)) = 52$ . Combined with the frontier value, we obtain the certified localization

$$tc_{\min}(\text{sycamore\_m20}) \in [53, 61.544]. \quad (\text{A.1})$$

402 On `reg3_250` the analogous certification is obstructed by the known hardness of certifying  
 403 high treewidth on expanders: the best machine-verified certificate (a 32-vertex minor of  $L(G)$ ,  
 404 checked by two independent exact solvers) gives  $tw(L(G)) \geq 18$  against a constructive upper  
 405 bound of 34.

406 The width bound has a ceiling:  $tc$  is a log-sum (1) over  $\sim |V|$  contractions, and the fron-  
 407 tier’s excess over the width bound—8.5 on `sycamore_m20`, close to the maximal possible  
 408  $\log_2 561 = 9.1$ —is invisible to any bound on a single contraction. Closing the interval (A.1)  
 409 from below requires *sum-form* bounds. For any tree, each internal node  $v$  with leaf set  $S_v$   
 410 satisfies  $\text{cost}_v \geq |\partial S_v|$ , the boundary of  $S_v$  in  $G$ . Writing  $b(k)$  for the isoperimetric profile of  $G$   
 411 (the minimal boundary over all  $k$ -vertex subsets), we prove:

412 **Theorem 1** (Sum-form profile bound). *For every contraction tree,  $tc \geq \log_2 F(n)$  where  $F$  obeys*  
 413 *the recursion  $F(1) = 0$ ,  $F(k) = 2^{b(k)} + \min_{1 \leq j \leq k/2} [F(j) + F(k-j)]$ .*

414 **Theorem 2** (Dyadic-window bound). *Descending from the root always into the larger child*  
 415 *visits, for every dyadic window  $W_j = (n/2^{j+1}, n/2^j]$ , at least one internal node whose leaf-set size*  
 416 *lies in  $W_j$ ; these nodes are distinct, hence  $tc \geq \log_2 \sum_j 2^{\min_{k \in W_j} b(k)}$ .*

417 Both theorems are validated computationally (exhaustively at small  $n$ , and on 3000 ran-  
 418 dom trees). Both fail to beat the max-form cap (Fig. 6): on `sycamore_m20` they reach 47.2  
 419 and 47.0 against the carving value 53. The reason is structural:

420 **Observation 1** (Profile bounds are capped at the bisection width). *Any lower bound computed*  
 421 *from the isoperimetric profile  $b(\cdot)$  alone is at most  $\max_k b(k) + \log_2 \log_2 n$ , and  $\max_k b(k)$  is*  
 422 *the bisection width—strictly below the balanced-cut (carving) value whenever the profile dips off-*  
 423 *center. On `sycamore_m20` it does: we exhibit a triply-verified 141-tensor subset with boundary*  
 424 *exactly 40, against temporal-slab boundaries of 53 (Fig. 7).*

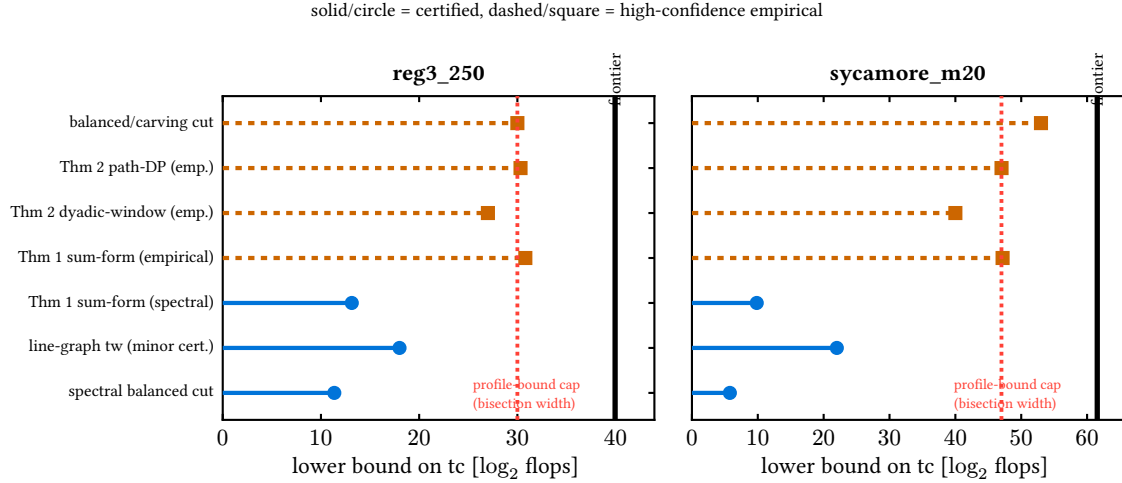


Figure 6: Lower-bound ladder for both closed instances. Solid stems with circles are certified (machine-checkable certificates); dashed stems with squares are high-confidence empirical values. Every profile-based bound (Theorems 1 and 2) is capped at the bisection width (dotted line, Observation 1); only the balanced/carving cut exceeds it. The solid vertical line is the optimizer frontier.

425 The  $47 \rightarrow 53$  gap lives in jointly-nested-cut structure that per-size minima discard; the  
 426  $53 \rightarrow 61.5$  gap is the log-sum residual. A complementary measurement (Fig. 8) shows why  
 427 search cannot exploit the residual either: reducing the largest contraction by one unit system-  
 428 atically multiplies the number of contractions just below the peak, so total flops behave as if  
 429 conserved under profile reshaping. This certified null fixes the design space for any challenger:  
 430 the gains must come from changing what is searched, not how.

## 431 B What does not work: falsified alternatives

432 Twenty further mechanisms were implemented, measured, and falsified under the same proto-  
 433 col, and their taxonomy (Table 3) is what makes the design of Sec. 5 non-arbitrary: *every mech-*  
 434 *anism that injected global information as search control failed; every mechanism that injected it*  
 435 *as input won*. Construction-time global structure (separators, spectral cuts, boundary combs)  
 436 loses to the greedy portfolio it seeds; search-control globality (targeting, tempering, blocking,  
 437 acceptance relaxation) breaks the annealer’s schedule invariants; exhaustive enumeration hits  
 438 certified combinatorial walls (window-exact optimality of the incumbent frontier; block counts  
 439 exploding past  $10^5$  within bits of the minimum boundary—with an outer-product counterex-  
 440 ample showing the connectivity-restricted variant is not even correct); and population policies  
 441 drain a small recombination reservoir (about ten productive recombinations) regardless of di-  
 442 versity management. Global cut surgery is, to our knowledge, the unique surviving design  
 443 point: it consumes global information (a better cut) as an *input* to a rebuild, leaving the an-  
 444 nealer’s control loop untouched.

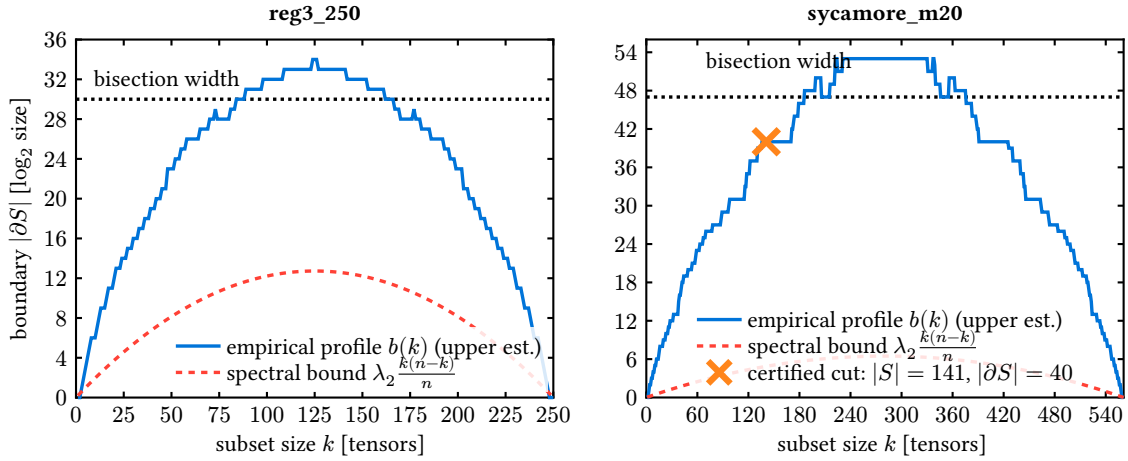


Figure 7: Isoperimetric profiles. The empirical profile  $b(k)$  (solid) dips well below the bisection width (dotted) at off-center sizes; the certified marker shows the 141-tensor sycamore\_m20 subset with boundary exactly 40. This dip is what caps all profile-based bounds (Observation 1).

## 445 C Supplementary figures

## 446 D Methodology: a research loop that distrusts itself

447 The results above were produced by an autonomous research loop—56 optimizer attempts  
 448 over ten cycles, each in an isolated worktree with a pre-registered hypothesis, scored by a  
 449 sandboxed validator—and the loop’s central design lesson is that its most important outputs  
 450 were *retractions* (Fig. 13). Three failure modes were caught by the harness rather than by  
 451 luck:

452 (i) *Schedule exploitation*. The first gate (a speed-versus-baseline bar) was passed by budget  
 453 hygiene alone—anytime writes and early termination—with no algorithmic content. The fix  
 454 was structural: fold the exploit into the reference, making schedule-only candidates score zero  
 455 by construction, and keep the exploit as a permanent negative control.

456 (ii) *Mis-tuned-baseline gains*. Eight mechanisms then “beat” the references by up to  $800\times$ —  
 457 until a strengthened reference (the one-line  $sc_{\text{target}}$  fix of Sec. 3) reclaimed the entire gain.  
 458 What survived was the tuning discovery itself, and the leaderboard practice of ratcheting refer-  
 459 ences to absorb every discovered improvement.

460 (iii) *Instrument bias*. A failed attempt’s diagnosis revealed that the independent scorer  
 461 silently rejected contraction trees deeper than  $\sim 475$  (a recursion limit), biasing the admissible  
 462 space against exactly the deep, MPS-like trees that suit circuit instances. The scorer was fixed,  
 463 its negative-control suite re-passed, and the affected hypothesis re-tested.

464 As the campaign scaled, the harness grew with it: parallel scoring lanes with per-child CPU  
 465 accounting, median-of-three scoring above 1000 tensors, early abort of hopeless confirmation  
 466 runs, and harness-clocked anytime curves that make time-to-frontier claims backdate-proof.  
 467 The pattern that generalizes: score only through a hardened independent validator; maintain  
 468 executable negative controls, and when a new exploit is found, patch the harness and add the  
 469 exploit to the control suite; ratchet references so yesterday’s wins are tomorrow’s baselines;  
 470 and report distributions wherever single runs are noisy. Under this regime, every claim in this  
 471 paper is either machine-certified, or labeled high-confidence with its evidence, or was retracted  
 472 before publication—and the audit trail (per-attempt logs, certificates, and cycle reports) ships  
 473 with the code.

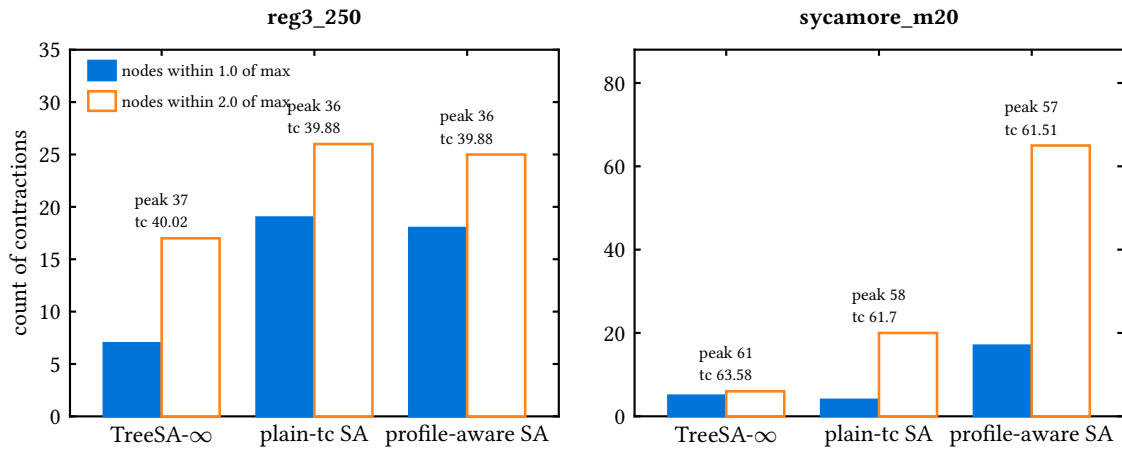


Figure 8: Cost-profile conservation at the frontier. Bars count contractions within 1.0 (solid) and 2.0 (outline) of the largest; labels give each tree’s peak cost and tc. Trees that lower the peak by one unit acquire several times more near-peak contractions at essentially unchanged tc.

## 474 E How the algorithm was found: steered autonomous research

475 The algorithm of this paper was produced by an autonomous research loop (machine-proposed  
 476 hypotheses, isolated implementations, hardened scoring) that was *steered* by a small number  
 477 of human decisions. (For LLM-driven *heuristic evolution* on this same problem—automated  
 478 search over code rather than a steered hypothesis loop—see [10].) Because the loop retained  
 479 every artifact, the causal path from the opening question to the algorithm can be reconstructed  
 480 exactly. Figure 14 shows that path left to right: the top row quotes the steering prompts, the  
 481 middle row shows only the attempts that carried results forward (most of the 56 attempts  
 482 served as controls, falsifications, or dead branches feeding Appendices B and D), and each  
 483 arrow states why the transition happened.

484 Three properties of this trajectory are worth stating explicitly. *First, every pivot was forced by*  
 485 *a measurement, not chosen freely.* The move from tc-optimization to the anytime axis happened  
 486 because thirteen mechanisms and a second toolchain had converged on one frontier that was  
 487 then certified locally optimal three independent ways; the move to scale happened because  
 488 time-to-frontier proved heavy-tailed at the claim threshold; the move to global-information  
 489 mechanisms happened because every search-control variant had failed with a structural cause  
 490 while every input-level variant had won. *Second, the human contributions were few but load-*  
 491 *bearing:* the demand for a strengthened reference (which produced the tuning result of Sec. 3),  
 492 the switch to real-provenance instances (which unlocked the simplification pass—the same  
 493 mechanism had been correctly measured as useless on the pre-simplified proxy two cycles  
 494 earlier), the question “can global information help the local search escape?” (which became  
 495 global cut surgery), and the directive to demonstrate the methods on inference workloads  
 496 (which produced the two-regime result of Sec. 7). *Third, retests are cheap when provenance*  
 497 *changes:* attempt 013’s null on the synthetic proxy and attempt 039’s records on the real circuit  
 498 are the same mechanism measured on different inputs, and only the pre-registered instance  
 499 provenance made the contradiction interpretable rather than embarrassing.

| family / mechanism (attempt)                               | measured cause of failure                                                                                                                                      |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Construction proxies</i>                                |                                                                                                                                                                |
| BFS/planar separators (037)                                | seeds lose to greedy portfolio even with separator injection                                                                                                   |
| spectral bisection (041)                                   | power iteration cannot converge (gap $\sim 1/n^2$ ); a perfect bisection is still dominated                                                                    |
| boundary-MPS combs (036/044)                               | cutwidth 90–126 lower-bounds any comb, far above balanced-tree records                                                                                         |
| mass randomized greedy (040)                               | quality ceiling 15–208 bits above frontier at 1.2 builds/s                                                                                                     |
| <i>Search-control globality</i>                            |                                                                                                                                                                |
| elimination-order (045)                                    | annealing flat landscape: 30 improvements per 2.5M moves; representation floor $\sim 2\times$ above tree space                                                 |
| congestion-softmax (042)                                   | targeting collapses onto the dominant node on peaky profiles; 40–80% dead traversal                                                                            |
| cache-blocked regional SA (043)                            | pointer trees have no memory contiguity; blocking starves global moves                                                                                         |
| replica-exchange tempering (048)                           | swap acceptance $\rightarrow 0\%$ at scale (energy gaps grow with $n$ )                                                                                        |
| late-acceptance hill climbing (051)                        | never cools on plateau-dominated landscapes; schedules are load-bearing                                                                                        |
| MCTS over prefixes (050)                                   | reaches depth 3 of $\sim 380$ at 8–25 rollouts/s                                                                                                               |
| <i>Exhaustive enumeration</i>                              |                                                                                                                                                                |
| window-exact splice (033)                                  | frontier trees are window-exact optimal ( $\leq 16$ leaves); the dominant window spans a constant fraction                                                     |
| Tamaki-style block DP (056)                                | feasible low-boundary blocks explode past $10^5$ – $10^6$ within bits of $\min_w$ ; outer-product counterexample falsifies the connectivity-restricted variant |
| <i>Population policy</i>                                   |                                                                                                                                                                |
| path-relinking / consensus / diversity pools (046/049/055) | basin complementarity drains after $\sim 10$ recombinations regardless of pool policy                                                                          |

Table 3: The falsified-mechanism taxonomy. Each row is an implemented, scored, and diagnosed attempt; attempt numbers index the public audit trail. The four families fail for four distinct measured reasons, and none of the failures is a tuning accident.

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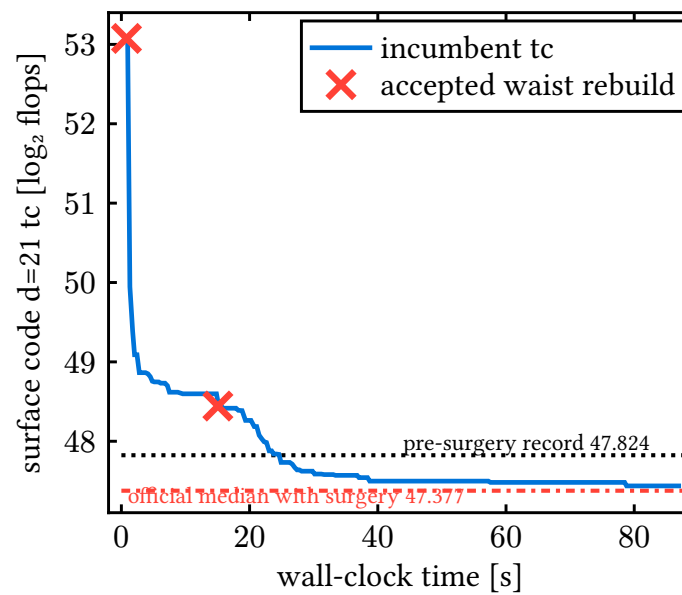


Figure 9: A representative surgery run on surface code  $d=21$ : incumbent tc (solid), accepted waist rebuilds (crosses), the pre-surgery record (dotted) and the official with-surgery median (dash-dotted). Each accepted rebuild moves the waist; surgery repeats at the new waist until no improving cut remains.

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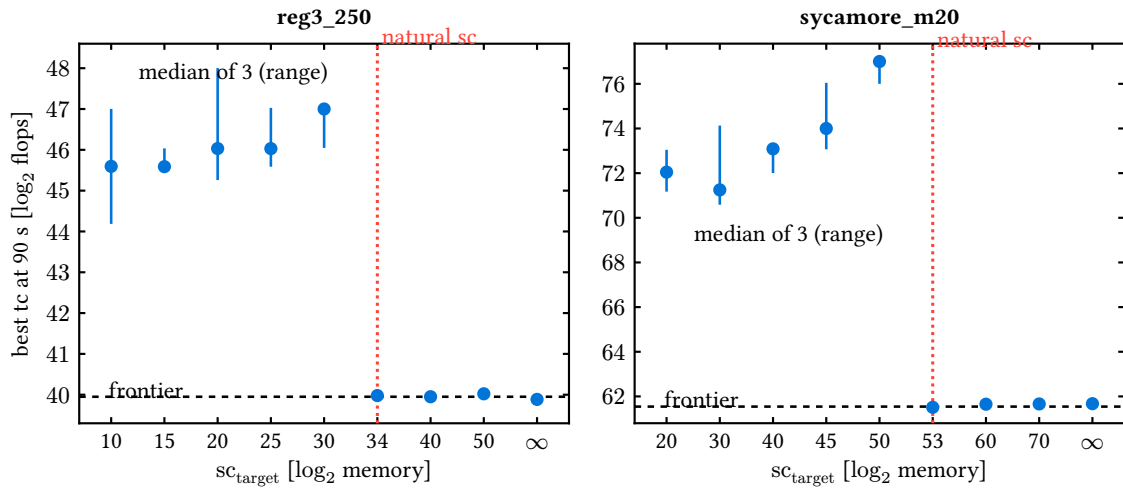


Figure 10: Best tc reached in 90 s of single-threaded annealing versus the memory target  $sc_{\text{target}}$  (median of three runs; whiskers span the range). The dashed line marks the converged frontier of Appendix A; the dotted vertical line marks the natural space cost of optimum-quality trees. The transition is a cliff at the natural value: every smaller target is comparably harmful, and every target at or above it, including  $sc_{\text{target}} = \infty$ , reaches the frontier. The shipped default is  $sc_{\text{target}} = 20$ .

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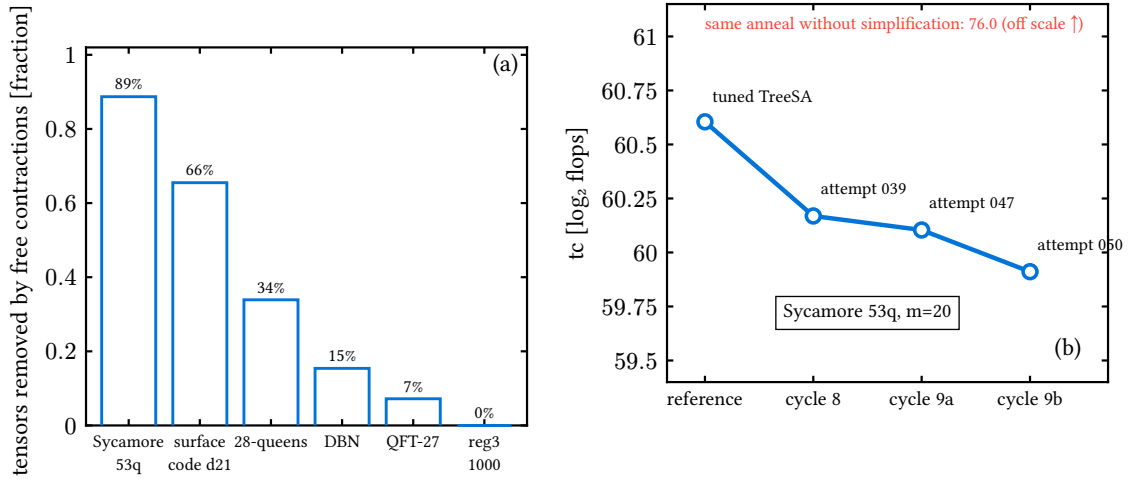


Figure 11: The preprocessing pass. (a) Fraction of tensors removed by the exact rank-non-increasing pass; the random-regular control is untouched at 0%. (b) The Sycamore 53q record march: the tuned reference (leftmost) and three successive simplify-then-anneal attempts. The same anneal without simplification reaches 76.0 (off scale): preprocessing, not search, carries this improvement.

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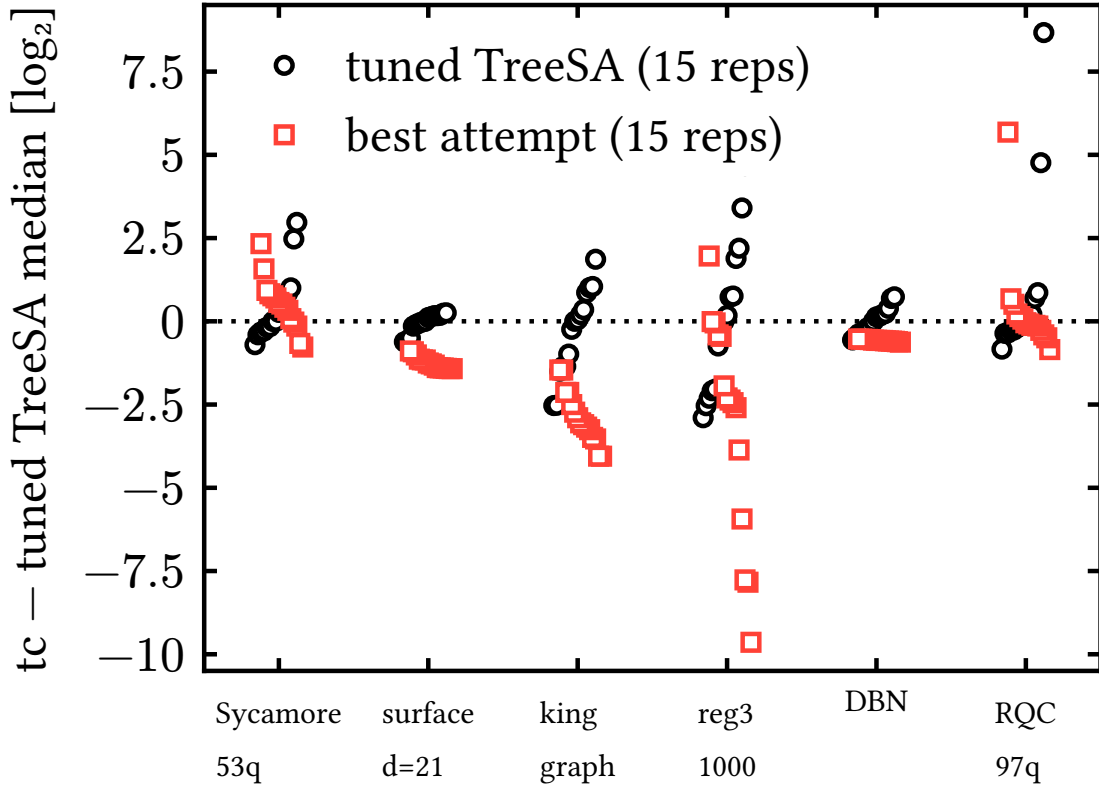


Figure 12: Run distributions of the same-machine campaign: all 15 repetitions per instance at 90 s, centered on the tuned reference’s median. Circles are the tuned reference, squares the best method per instance.

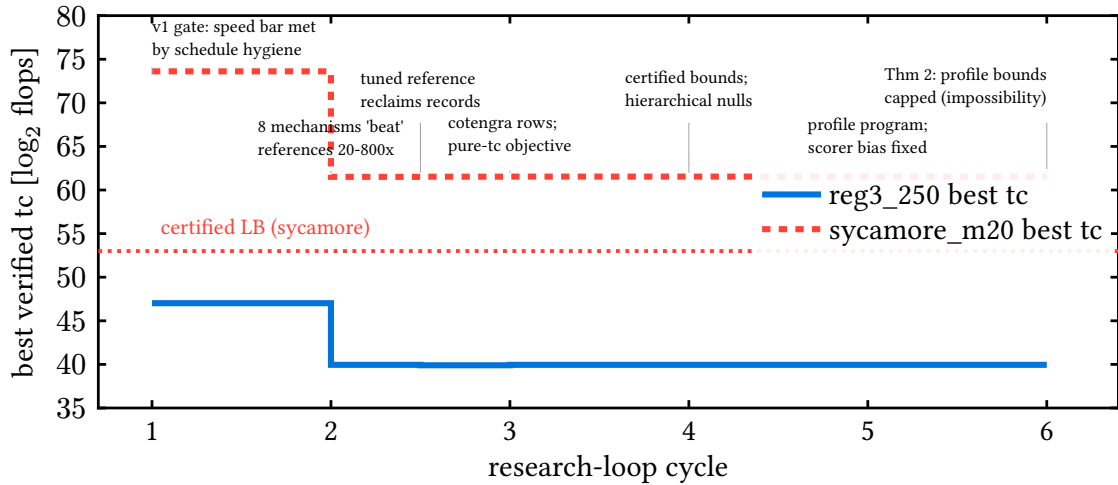


Figure 13: Best verified tc over the research-loop cycles. Apparent progress in cycle 2 is reclaimed by the strengthened reference (cycle 2.5); thereafter the closed-instance frontier is flat while the certified picture grows, and cycles 7–10 move the record board on the large structured instances instead. Annotations mark the three instrument-caught failure modes.

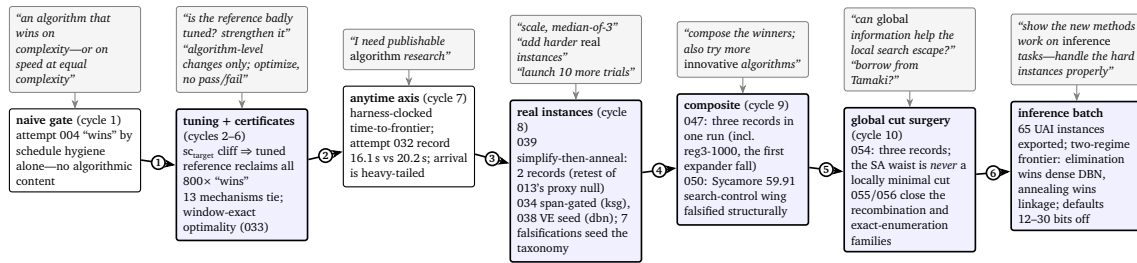


Figure 14: The research roadmap, left to right. Top row (grey): the human steering prompts, essentially verbatim. Bottom row: the result-carrying attempts only; most of the 56 attempts are controls and falsifications that feed Table 3 and Appendix D. Each numbered arrow is a transition forced by a measurement: (1) the gate was passed without an algorithm, so the gate was hardened and references ratcheted; (2) tc saturated and was certified locally optimal, so the objective axis changed to arrival time; (3) speed gains sat at the confirmation noise floor while scale budgets were non-converged, so the regime changed; (4) the winning mechanisms proved complementary, so they were composed while exploration continued; (5) every win injected global information as *input* while every search-control variant failed, so global information was aimed directly at the local search's blind spot—the dominant cut; (6) the mechanism portfolio was complete, so it was pointed at a second application domain to test transfer.